



Proton NMR Spectroscopy

Video Series 2

Video 3: Complex Splitting Patterns





Video Series Goals

Video 1: NMR Spectra of Mixtures

- Diagnostic NMR signals
- Calculating Percent Composition of a Mixture

Video 2: NMR Solvents

- Deuterated Solvents
- Solvent Signals in NMR

Video 3: Splitting Pattern Complexity (**Current**)

- Extension of “n+1” Rule

Video 4: Long Range Coupling

- 4-Bond (4J) Coupling

Video 5: Diastereotopic Protons

- 2-Bond (2J) Coupling



Multiple Coupling Partners

When protons have multiple coupling partners, the splitting patterns can become complicated if the **coupling constants are different** for each interaction.

Generalized “n+1” Rule:

- **n** neighboring protons with **the same coupling constant (J)** will split a signal into **n+1** peaks.
- Group protons **with the same J**, and **apply the n+1 rule to each group sequentially**

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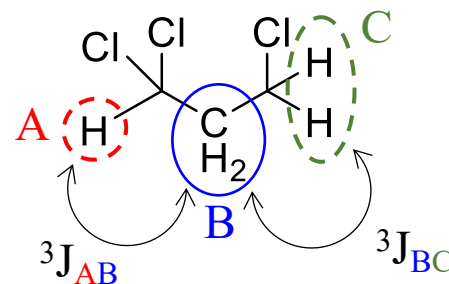
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What is the splitting pattern of signal **B**?

Consider three theoretical possibilities:

1. $^3J_{AB} = ^3J_{BC}$
2. $^3J_{AB} > ^3J_{BC}$
3. $^3J_{AB} < ^3J_{BC}$



Tree Diagrams and Splitting Patterns

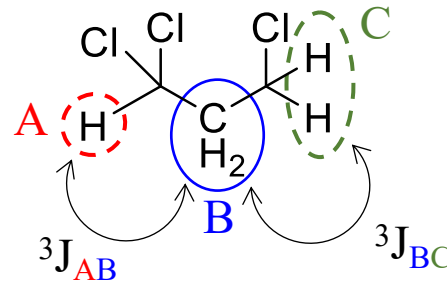
Tree Diagrams

- **J** is the distance between split peaks in **Hz**
- Use the **n+1** rule sequentially for every coupling interaction

What is the splitting pattern of signal **B**?

If ${}^3J_{AB} = {}^3J_{BC} = 6 \text{ Hz}$

Correctly drawn tree diagrams will always predict the correct splitting pattern
The generalized n+1 rule is a shortcut we can use to avoid drawing tree diagrams.



B
|

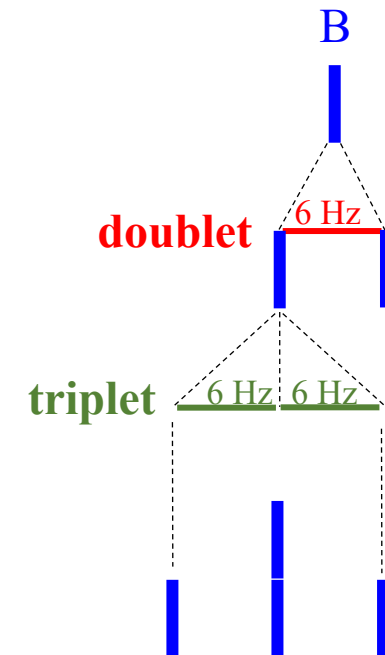
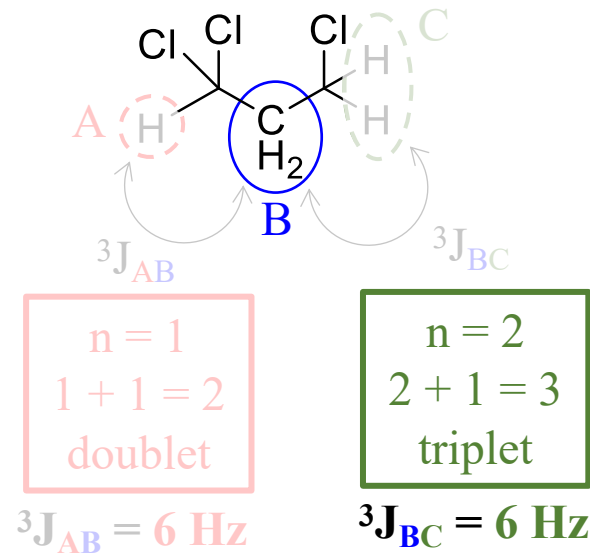
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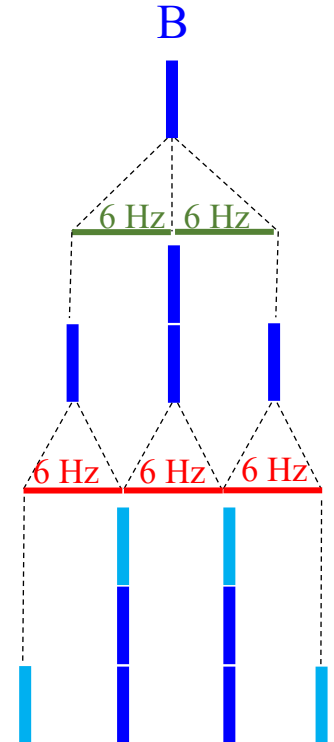
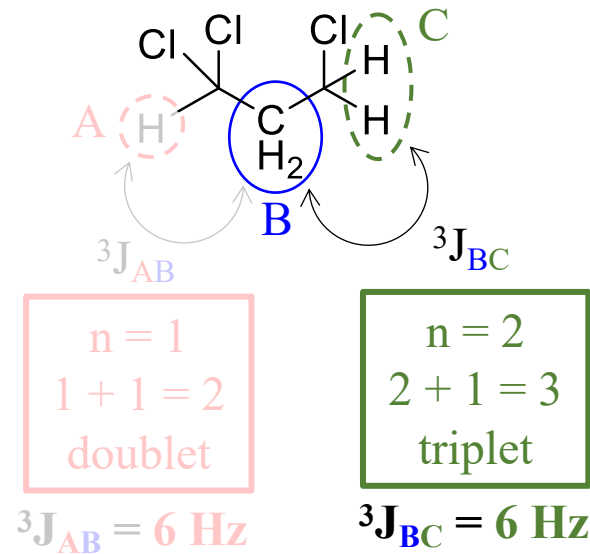
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Tree Diagrams and Splitting Patterns

Tree Diagrams

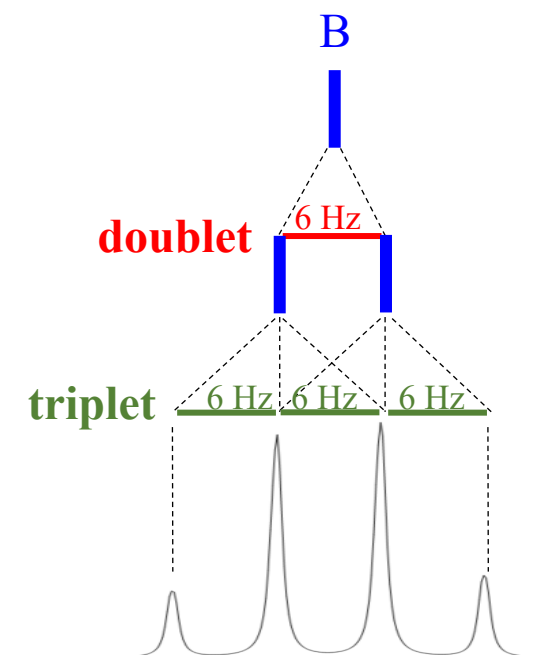
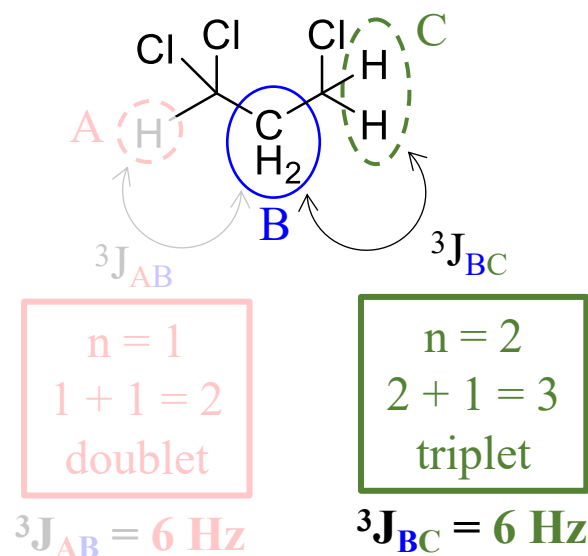
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Generalized “n+1” Rule:

- **n** neighboring protons with **the same coupling constant (J)** will split a signal into **n+1** peaks.
- Group protons with **the same J**, and **apply the n+1 rule to each group sequentially**



Looks like a **quartet!**

J values are all equal

One “group” of protons

Use the n+1 rule on each group

Tree Diagrams and Splitting Patterns

Tree Diagrams

- **J** is the distance between split peaks in **Hz**
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What is the splitting pattern of signal **B**?

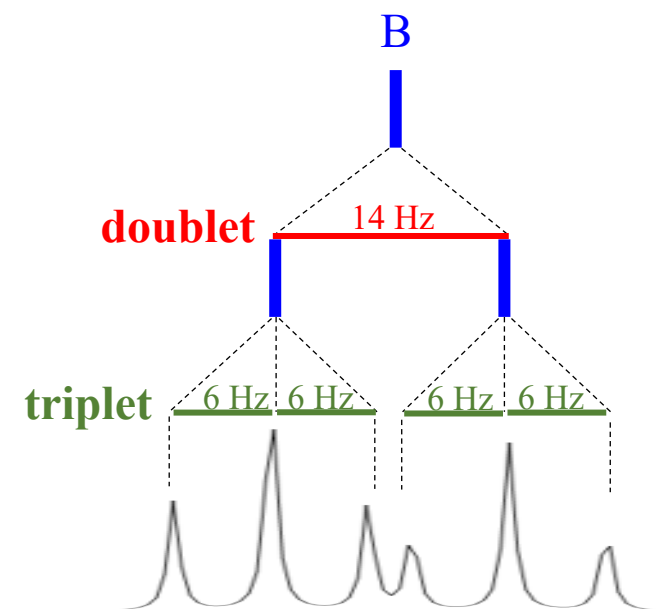
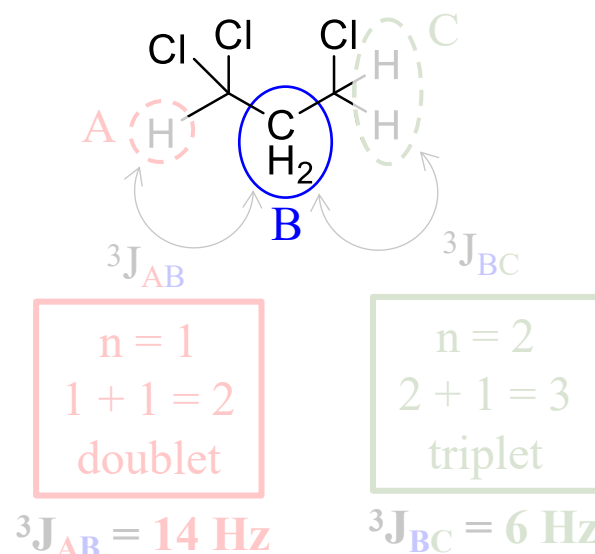
If ${}^3J_{AB} > {}^3J_{BC}$

${}^3J_{AB} = 14 \text{ Hz}$

${}^3J_{BC} = 6 \text{ Hz}$

Generalized “n+1” Rule:

- **n** neighboring protons with **the same coupling constant (J)** will split a signal into **n+1** peaks.
- Group protons with **the same J**, and **apply the n+1 rule to each group sequentially**



Doublet of triplets (dt)

J values are not equal

Two groups of protons

Use n+1 rule twice

Tree Diagrams and Splitting Patterns

Tree Diagrams

- **J** is the distance between split peaks in **Hz**
- Use the **n+1** rule sequentially for each coupling interaction

What is the splitting pattern of signal **B**?

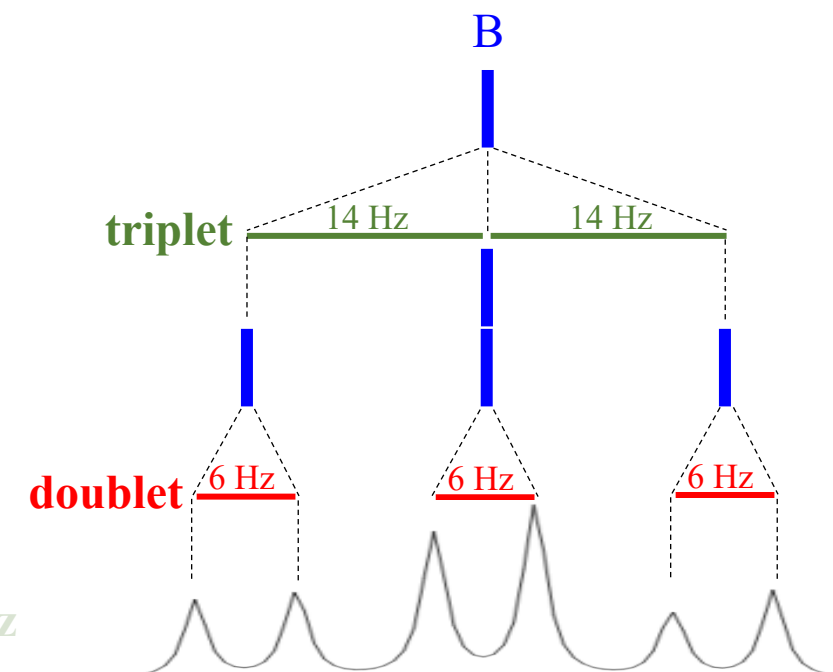
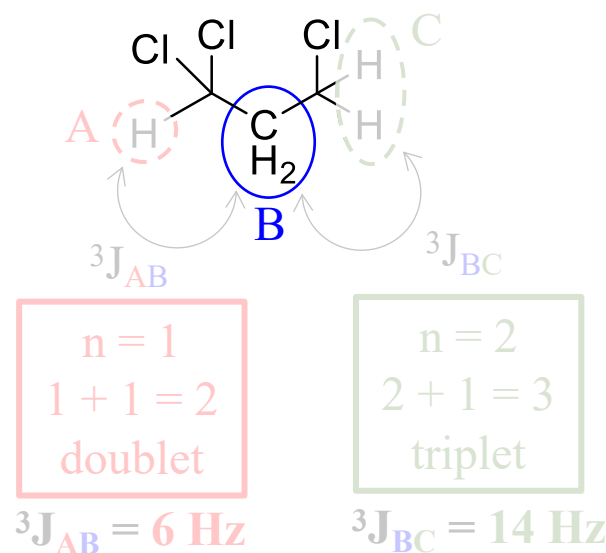
If ${}^3J_{AB} < {}^3J_{BC}$

${}^3J_{AB} = 6 \text{ Hz}$

${}^3J_{BC} = 14 \text{ Hz}$

Generalized “n+1” Rule:

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Triplet of doublets (td)

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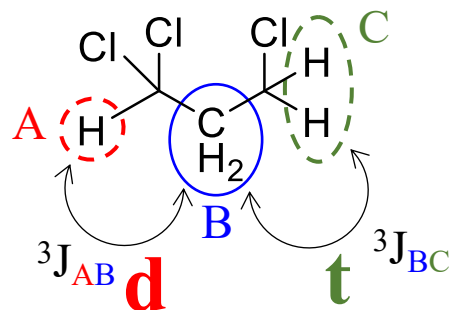
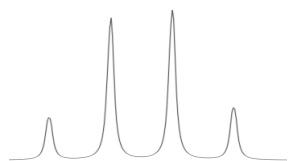
Multiple Coupling Partners

If a set of protons has multiple coupling partners:

Are **all J values the same**? Group all coupled protons and use the **n+1 rule** once.

Are the **J values different**? Use the **n+1 rule sequentially** for each group of protons.

If ${}^3J_{AB} = {}^3J_{BC}$
 $n = 3$
 $n + 1 = 4$
quartet

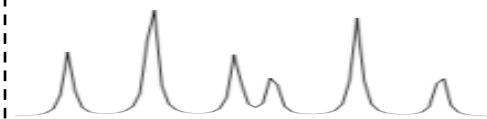


If ${}^3J_{AB} \neq {}^3J_{BC}$

$n = 1$
 $n + 1 = 2$ (d)

$n = 2$
 $n + 1 = 3$ (t)

dt (doublet is larger)



${}^3J_{AB} > {}^3J_{BC}$

td (triplet is larger)



${}^3J_{AB} < {}^3J_{BC}$

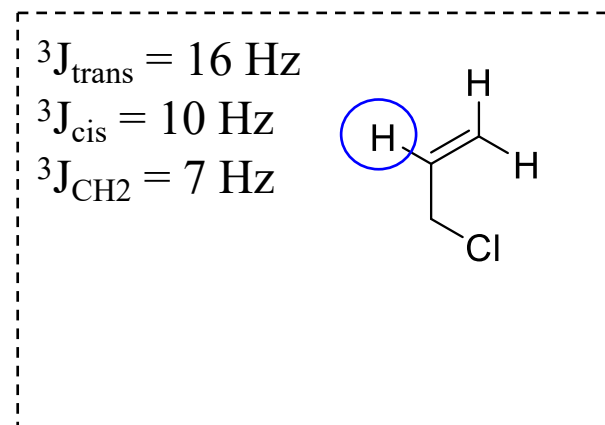
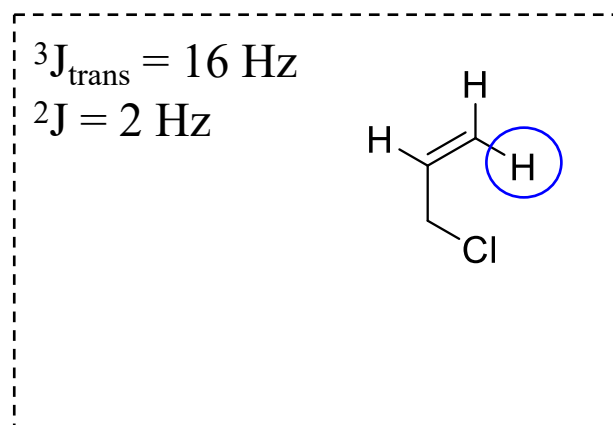
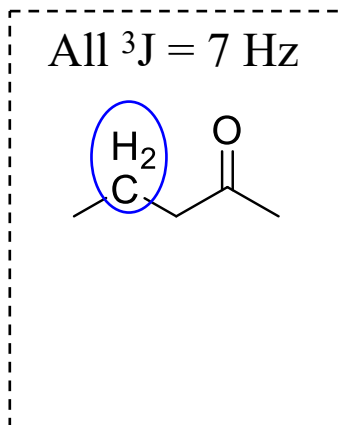
Predicting Splitting Patterns

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Predict the splitting pattern of the hydrogens circled in **blue** for each molecule (e.g. td, d, q)



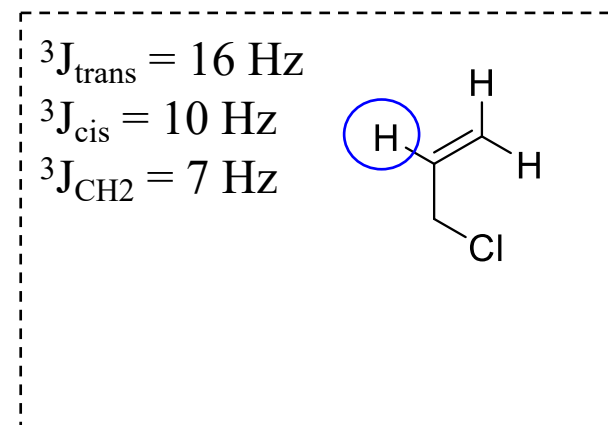
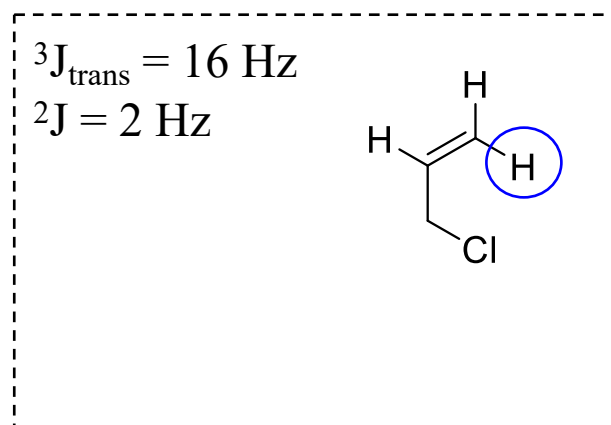
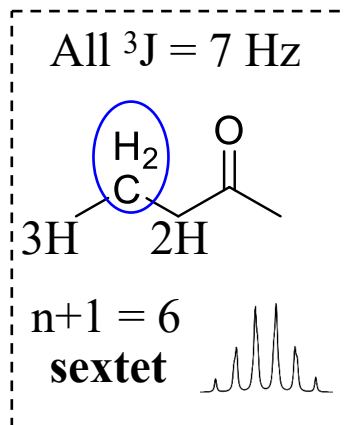
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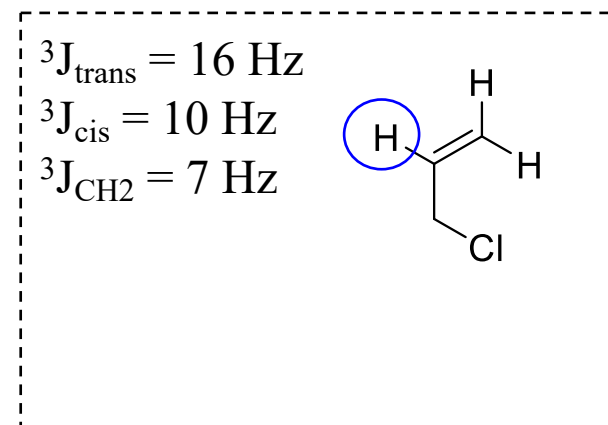
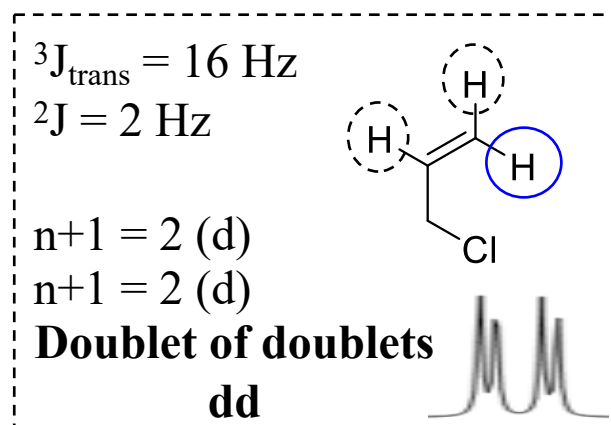
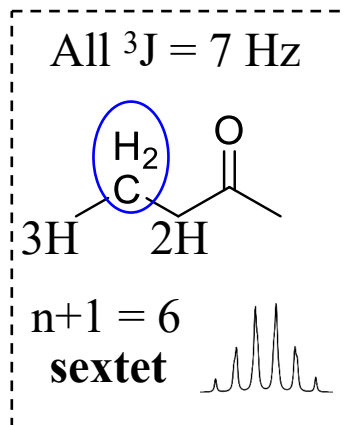
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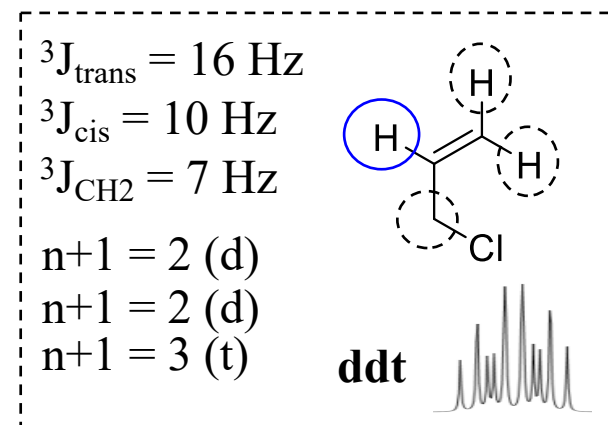
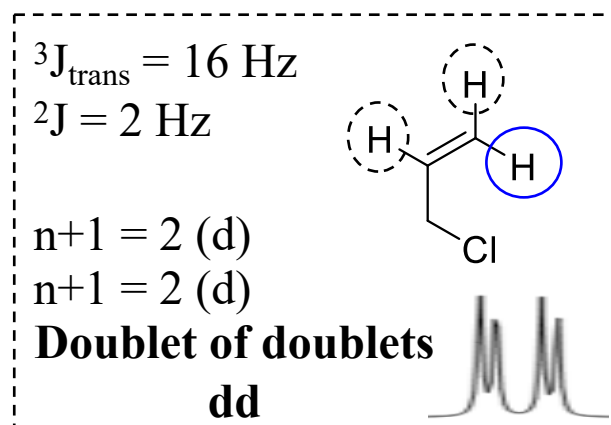
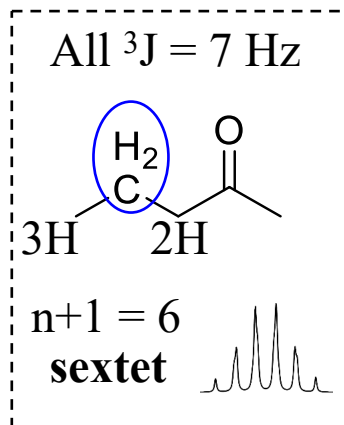
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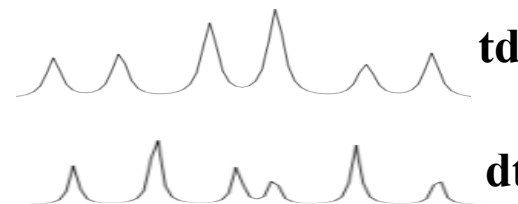
Video Summary

If proton has multiple coupling partners, group any protons with the same **J**

- Are **all J values the same**? Group all coupled protons and use the **n+1 rule** once.
- Are the **J values different**? Use the **n+1 rule sequentially** for each group of protons.

When naming a complex splitting pattern, start with the largest coupling constant

- A triplet of doublets (**td**) will have a larger triplet coupling constant
- A doublet of triplets (**dt**) will have a larger doublet coupling constant



Next video: Long Range (4J) Coupling